Spec → Code: ONC_LSN_003_JR_STD

Exported 8/28/2026, 8:01:32 PM · Tenant: default

1. Spec

REPORT ID

ONC_LSN_003_JR_STD

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

Missing Lesion Evaluation. This report will list subjects who are missing a Lesion Evaluation for each lesion every specific interval of time following the randomization date/start date of the corresponding treatment phase.

COLUMN HEADING

Table 1
1. Country
2. Site
3. Subject Number
4. Visit
5. Date of Visit
6. Form
7. Seq
8. Lesion Identification
9. Expected Timeframe
10. Criteria Value1
11. Criteria Value2
12. Criteria Value3

Table 2
1. Country
2. Site
3. Subject Number
4. Starting Point
5. Starting Point Date
6. Week
7. Form
8. Seq
9. Lesion Identification
10. Expected Timeframe
11. Criteria Value1
12. Criteria Value2
13. Criteria Value3
14. Min cut off form
15. Min cut off seq
16. Min cut off date
17. OEDRA Date
18. PMS1A Date

Forms/Items:
Table 1
1. SA. Country
2. SA. Site Mnemonic
3. SA. Subject Number
4. PMS1/PMST. Visit Mnemonic
5. DOV.Date of Visit where PMS1/PMST is present
6. PMS1/PMST.Form Mnemonic

7. Lesion Evaluation Form.Formidx
8. Lesion Evaluation Form.Lesion Identification/Tumor Identification
9. As specified by Note 20
10. As specified by Note 31
11. As specified by Note 31
12. As specified by Note 31

Table 2
1. SA. Country
2. SA. Site Mnemonic
3. SA. Subject Number
4. First Visitmnemonic of Treatment Phase.Visit Mnemonic
5. First Visitmnemonic of Treatment Phase.DOV.Date of Visit
6. As specified by Note 21
7. Lesion Evaluation Form.Form Mnemonic
8. Lesion Evaluation Form.Formidx
9. Lesion Evaluation Form.Lesion Identification/Tumor Identification
10. As specified by Note 20
11. As specified by Note 31
12. As specified by Note 31
13. As specified by Note 31
14. As specified by Note 33
15. As specified by Note 34
16. As specified by Note 35
17. As specified by Note 37
18. As specified by Note 38

LOGIC

Prior to processing the logic, apply condition(s) according to the supporting doc to check if subjects match the criterial or not.

"Criteria Form1"." Criteria Itemquestion1" = "Criteria Value1" AND
"Criteria Form2"." Criteria Itemquestion2" = "Criteria Value2" AND
"Criteria Form3"." Criteria Itemquestion3" = "Criteria Value3" AND

Logic 1.
Logic 1.1 There is a 'PMS Form' record with Trial Segment Reference = 'Trial Segment Reference'
AND
Logic 1.2 There is no LSN eCRF."DATE OF LESION EVALUATION" for each LSN records that belong to the corresponding treatment phase meet below criteria:
Logic 1.3 LSN eCRF."DATE OF LESION EVALUATION" >= DOV. Date of Visit where 'PMS Form' is located - 'Lower window of days'
AND
Logic 1.4 LSN eCRF."DATE OF LESION EVALUATION" <= DOV. Date of Visit where 'PMS Form' is located + 'Upper window of days'
AND
Logic 1.5 LSN."Method of Assessment" = 'Method of Assessment' per supporting document for the corresponding 'Group for Method'
AND
(
Logic 1.6 There is not PMS1A record with Trial Segment Reference = 'PMS1A Trial Segment Reference' AND Date of Assessment <= DOV.Date of Visit where 'PMS Form' is located - 'Lower window of days'
AND
Logic 1.7 There is not OEDRA record with Date of Assessment <= DOV.Date of Visit where 'PMS Form' is located - 'Lower window of days'

)

Logic 2
Logic 2.1 System date > 'Expected Timeframe'
AND
Logic 2.2 There is not at least one LSN record in the add entry section meet the below criteria:
(
Logic 2.3 LSN eCRF."DATE OF LESION EVALUATION" within the 'Expected Timeframe' for each LSN record corresponding to each treatment phase
AND
Logic 2.4 LSN eCRF."METHOD OF ASSESSMENT" = 'Method of Assessment' per supporting document according to the corresponding 'Group for Method'
)
AND
(
Logic 2.5 LSN eCRF."LESION NUMBER" is not present in other record on the LSN eCRF."INDICATE IF LESION MERGED OR SPLIT"
OR
Logic 2.6 LSN eCRF."LESION NUMBER" is present in other LSN eCRF."INDICATE IF LESION MERGED OR SPLIT but 'Expected Timeframe' is before to LSN eCRF."DATE OF LESION EVALUATION"
AND
Logic 2.7 LSN eCRF."INDICATE IF LESION MERGED OR SPLIT" is NULL
OR
Logic 2.8 LSN eCRF."INDICATE IF LESION MERGED OR SPLIT" is NOT NULL AND missing "Date" is after the LSN eCRF."DATE OF INITIAL IDENTIFICATION"
)
AND
Logic 2.9 There is not at least one record on the 'Oncologic Treatment' form with ODBF."START DATE OF REGIMEN"/ "START DATE OF MEDICATION". <> NULL OR 'Expected Timeframe' is before the minimum ODBF."START DATE OF REGIMEN"/ " START DATE OF MEDICATION"Date entered across all records present in the form
AND
Logic 2.10 There is not at least one record on the 'Oncologic Radiation' form with ORAD2.Date <> NULL AND ORAD2.Reason <> Palliative OR 'Expected Timeframe' is before the minimum ORAD2.Date entered across all records present in the form
AND
Logic 2.11 There is not at least one record on the 'Oncologic Surgery' form with OSOF eCRF." "DATE OF SURGERY/ DATE OF SURGERY / DATE OF SURGERY <> NULL OR 'Expected Timeframe' is before the minimum OSOF eCRF." "DATE OF SURGERY/ DATE OF SURGERY / DATE OF SURGERY entered across all records present in the form
AND
(
Logic 2.12 There is not a PMS1A record with Trial Segment Reference = 'PMS1A Trial Segment Reference' AND Date of Assessment <= 'Expected Timeframe'
OR
Logic 2.13 There is not an OEDRA record with Date of Assessment <= 'Expected Timeframe'
)
AND
(
Logic 2.14 The subsequent treatment phase 'Minimal start timepoint' (DOV.Date of Visit OR Date of Assessment of SR) = NULL
OR
Logic 2.15 'Expected Timeframe' < 'Minimal start timepoint' (DOV.Date of Visit OR Date of Assessment of SR) of subsequent treatment phase
)
AND

Logic 2.16 There is NOT at least one RNT record with RNT.Date of Assessment <> NULL

Notes:
1- 'Lesion Evaluation Form', 'Lower window of days', 'Upper window of days', 'Trial Segment Reference', 'PMS1A Trial Segment Reference', 'Minimal start timepoint', 'Oncologic Treatment', 'Oncologic Radiation', 'Oncologic Surgery', 'Method of Asssessment' and 'Group for Method' are values from supporting document
2- Equivalent items questions across different LSN eCRFs are going to be retrieved from the 'LSN_supp_doc' global supporting document.
3- If 'Upper window of days' is entered as NA in supporting document, then report will ignore Logic 1.4
4- 'PMS Form' = PMS1/PMST
5- For Logic 1.6 and Logic 1.7 Logic 2.12 and Logic 2.13: PMS1A is used in older studies while OEDRA is used in newer studies, therefore 'PMS1A Trial Segment Reference' could be entered as NA in supporting document.
6- LSN eCRF."DATE OF LESION EVALUATION", LSN eCRF."DATE OF INITIAL IDENTIFICATION", LSN eCRF."INDICATE IF LESION MERGED OR SPLIT", LSN eCRF." TUMOR IDENTIFICATION", LSN eCRF."LESION NUMBER", LSN eCRF."METHOD OF ASSESSMENT" are going to be retrieved from the 'LSN_supp_doc' global supporting document column C
7- ODBF."START DATE OF REGIMEN"/ "START DATE OF MEDICATION". are going to be retrieved from the 'ODBF_supp_doc' global supporting document column C
8- ORAD2.Date = Start Date
9- ORAD2.Reason <> Palliative refers to the item 'Reason for Radiation' <> 'Palliative Treatment or Symptom Control' OR 'Palliative Treatment or Symptom Control of Metastatic Disease' for older studies and 'Treatment Category' <> 'Palliative' for newer studies.
10- OSOF. "DATE OF SURGERY" are going to be retrieved from the 'OSOF_supp_doc' global supporting document column C
11- Report will work for randomized patients only
12- Report will work when LSN ECRF has been initialized. If the form is empty, then, no output is produced.
13- Erroneous visits and deleted records will not be considered for programming
14- For ORAD2.Start date or OSOF.Date and Time Performed or ODBF.Start Date of Regime/START DATE OF MEDICATION are entered as UNK, day UNK will use the last day of the month, month UNK will use the last month of the year.
15- If 'Oncologic Treatment' from supporting document is entered as NA, then the report will ignore Logic 2.9
16- If 'Oncologic Radiation' from supporting document is entered as NA, then the report will ignore Logic 2.10
17- If 'Oncologic Surgery' from supporting document is entered as NA, then the report will ignore Logic 2.11
18- If a date is displayed in the output the report will display it in format 'DDMMMYYYY'
19- Logic 1 will check missing LSN eCRF."DATE OF LESION EVALUATION" for each lesion record
20- For table 1 column 9/table 2 column 10 the report will display the expected timeframe in format 'DDMMMYYYY to DDMMMYYYY' e.g., '01JAN2020 to 15JAN2020'
21- For table 2 and column 6 the report will display the result of 'Start week' + N*Frequency. Where N starts at 0 and increments by 1 for each expected timeframe.
22- LSN eCRF."INDICATE IF LESION MERGED OR SPLIT" is not available for older trials, therefore, the report will not consider this item when it is not present.
23- For Scenario 2 the 'stop week' can be entered as NA if per protocol imaging assessments should be performed until disease progression, death, withdrawal, the end of the study etc and does not truly mention a real 'stop week'. For these cases the report will consider 'stop week' as system date and will continue output missing lesion evaluations until PMS1A/OEDRA Date of Assessment is less than 'Expected Timeframe' according to the Logic 2.12 and Logic 2.13.
24- 'Expected Timeframe' is always expected to be after LSN eCRF."DATE OF INITIAL IDENTIFICATION". There might be cases where a new lesion appears after cycle 1, therefore, do not output if any missing 'Expected Timeframe' in LSN eCRF."DATE OF LESION EVALUATION" is before LSN eCRF."DATE OF INITIAL IDENTIFICATION".
25- 'Expected Timeframe' will be calculated using the items 'Minimal start timepoint', 'Start week', 'Stop week', 'Frequency', 'Lower window of days', 'Upper window of days'.
Steps to calculate the 'Expected Timeframe':
A. Calculate different timepoints by doing the following mathematical operation:
Timepoint 1: Minimal start timepoint + Start week

Timepoint 2: Minimal start timepoint + Start week + 1*Frequency
Timepoint 3: Minimal start timepoint + Start week + 2*Frequency
Timepoint N: Minimal start timepoint + Start week + (N-1)*Frequency
B. Timepoints calculated on step A should be: Minimal start timepoint + Start week <= Timepoint N <= Minimal start timepoint + Stop week
C. Expected Timeframe = From Timepoint N - 'Lower window of days' to Timepoint N + 'Upper window of days'
Example:
A. Minimal start timepoint = 01-Jan-2020
B. Start week = 6
C. Stop week = 30
D. Frequency = 6
E. Lower window of days = 7
F. Upper window of days = 7
Timepoint 1 = 01-Jan-2020 + 6 weeks = 12-Feb-2020
Expected Timeframe 1 = From 05-Feb-2020 to 19-Feb-2020
Timepoint 2 = 01-Jan-2020 + 6 weeks + 6 weeks = 25-Mar-2020
Expected Timeframe 2 = From 18-Mar-2020 to 01-Apr-2020
Timepoint 3 = 01-Jan-2020 + 6 weeks + 12 weeks = 06-May-2020
Expected Timeframe 3 = From 29-Apr-2020 to 13-May-2020
26- For Logic 2.1, Logic 2.4 and Logic 2.13 use the maximum date of the range for calculation e.g., if the range is "From 01-Jun-2020 to 10-Jun-2020" use 10-Jun-2020. For Logic 2.7, Logic 2.8, Logic 2.9, Logic2.10, Logic 2.11 use the minimum date e.g., use 01-Jun-2020.
27- For Logic 2, if Expected Timeframe of treatment phase X overlaps with start date of treatment phase X+1 then no output should be produced for treatment phase X. For example: if phase 2 starts on 14Feb2018, '45 week' check for phase 1 is on 10FEB2018 and acceptable timeframe is 03FEB2018~17FEB2018 (lower and upper window is 7 days) then no output is produced for '45 week'.
28- If supporting doc is not provided, display the message as "ONC_LSN_003_JR_STD Supporting Document Not Present for This Trial"
Requires supporting document to be named ONC_LSN_003_supp_<MKxxxxxxx>.csv with columns and content as described (Criteria Form1/2/3, Criteria Itemquestion1/2/3, Criteria Value1/2/3, Phase Order, Minimal start timepoint, Lesion Evaluation Form, Trial Segment Reference, PMS1A Trial Segment Reference, Group for Method, Method of Assessment, Lower window of days, Upper window of days, Frequency, Start week, Stop week, Oncologic Treatment, Oncologic Radiation, Oncologic Surgery, Scenario) - full column definitions and multiple examples for Scenario 1 and Scenario 2 provided in source text.
29- The supporting document is required for all trials if treatment phase is applied to logic, even if only one treatment phase is present. This supporting document is only required to be created once per trial. If it was already created and uploaded, there is no need to duplicate it. The supporting document should be named as study_phase_supp_<MKxxxxxxx>.csv with columns: Phase Order (required), Treatment Phase (name), Minimal start timepoint (required, first visit on which treatment is expected to be administered on each phase). Example:
1, Initial Treatment Phase, Cycle 1
2, Second Course Phase, Cycle 1 Course 2
3, Crossover Phase, Crossover Cycle 1
The actual information provided by the supporting doc can be outputted by selecting Y in the parameter "Study Phase Page".
30- The report logic should be checked within the same treatment phase defined in the study phase supporting document.
For example, if user provides two Treatment Phases in supporting document like below:
Phase Order, Treatment phase, Minimal start timepoint
1,Initial Treatment Phase,Cycle 1
2,Second Course Phase,Second Course Cycle 1

The treatment phase will be identified as following for trials where LSN eCRF."INDICATE IF LESION MERGED OR SPLIT" is present:
Initial Treatment Phase: LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 1 <= Cycle 1 <= LSN

eCRF."DATE OF INITIAL IDENTIFICATION". with condition 2 OR LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 3 < Second Course Cycle 1
Second Course Phase: LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 1 <= Second Course Cycle 1 <= LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 2 OR LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 3

The treatment phase will be identified as following for trials where Indicate if Lesion Merged or Split is NOT present:
Initial Treatment Phase: LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 4 <= Cycle 1 <= LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 3 < Second Course Cycle 1
Second Course Phase: LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 4 <= Second Course Cycle 1 <= LSN eCRF."DATE OF INITIAL IDENTIFICATION". with condition 3

Condition 1 = LSN eCRF." TUMOR IDENTIFICATION"= 'non-target/target' and Indicate if Lesion Merged or Split = NULL
Condition 2 = LSN eCRF." TUMOR IDENTIFICATION"= 'non-target/target' and Indicate if Lesion Merged or Split <> NULL
Condition 3 = LSN eCRF." TUMOR IDENTIFICATION"= 'new'
Condition 4 = LSN eCRF." TUMOR IDENTIFICATION"= 'non-target/target'

'non-target/target' = Non-Target, Target, Target Extra Nodal, Target Nodal, Non-Target Nodal, Non-Target Extra Nodal, Non-Target Enhancing, Non-Target Non-Enhancing, Bone Lesion

'new' = New, New Lesion Since Last Evaluation, New Non-Target Extra Nodal, New Non-Target Nodal, New Target Nodal, New Target Extra Nodal, New Bone Lesion

31. Report will display Criteria Value as specified in supp document in the corresponding columns Criteria Value1, Criteria Value2 and Criteria Value3. These Criteria Value is the value from eCRF specified in Column 'Criteria Form1/2/3' and in Item specified in column 'Criteria Itemquestion1/2/3'.
32. Report will use different settings for different Criteria Values specified in supporting document
33. For table 2 and column 12 the report will display the form that have min date for Logic 2.9~2.13.
34. For table 2 and column 13 the report will display the seq that have min date for Logic 2.9~2.13.
35. For table 2 and column 14 the report will display the min date for Logic 2.9~2.13 in format 'YYYY/MM/DD' e.g., '2020/09/UNK'.
36. Report check PMS1A record within the same phase order if multiple PMS1A/OEDRA Trial Segment Reference values are provided for one phase order
37. For Table 2 and Column 15, display the date from OEDRA.Date of Assessment in format 'DDMMMYYYY'. If not available, then display blank
38. For Table 2 and Column 16, display the date from PMS1A.Date of Assessment in format 'DDMMMYYYY'. If not available, then display blank
39. If Group for Method and Method of Assessment are entered as NA in the supporting document, the report will ignore Logic 1.5 and 2.4
40. The "Other" free textbox in the "Method of Assessment" field will not be considered for programming
41. RNT is not used in old trials, only new trials have implemented RNT, therefore, Logic 2.16 is not applicable for old trials
42. The form listed in 'Cut-off Exclude Form' parameter will not be taken into consideration in cut-off calculation in Logic 1 and Logic2. For example, if 'PMS1A' is entered for 'Cut-off Exclude Form' parameter, logic 1.6 and logic 2.12 will be ignored

Prompts (runtime parameters):
Country, First Site, Last Site, First Subject, Last Subject, List of Subjects, Supporting Doc Page (N/Y, Default as N), Study Phase Page (N/Y, default as N), Lesion Evaluation Form, Cut-off Exclude Form

Grouping:

Table 1 will show output when Logic 1 is satisfied
Table 2 will show output when Logic 2 is satisfied

REPORT SCENARIOS

Scenario 1 Output Yes:
A. There is no LSN eCRF."DATE OF LESION EVALUATION" present for each LSN record that belongs to each treatment phase in the expected window of days from discontinuation of that treatment phase and with the corresponding LSN eCRF."METHOD OF ASSESSMENT"

Scenario 1 Output No
A. There is at least one LSN eCRF."DATE OF LESION EVALUATION" present for each LSN record that belongs to each treatment phase in the expected window of days from discontinuation of that treatment phase and with the corresponding LSN eCRF."METHOD OF ASSESSMENT"
B. There is no LSN eCRF."DATE OF LESION EVALUATION" present for each LSN record that belongs to each treatment phase in the expected window of days from discontinuation of that treatment phase but discontinuation visit is erroneous
C. Patient is not randomized
D. LSN form is empty

Scenario 2 Output Yes:
A. There is no at least one LSN eCRF."DATE OF LESION EVALUATION" for each expected timeframe according to the start/stop week and frequency from cycle 1 of each treatment phase and for each LSN record that belong to each treatment phase and with the corresponding LSN eCRF."METHOD OF ASSESSMENT"

Scenario 2 Output No
A. There is no at least one LSN eCRF."DATE OF LESION EVALUATION" for each expected timeframe according to the start/stop week and frequency from cycle 1 of each treatment phase and with the corresponding LSN eCRF."METHOD OF ASSESSMENT" for each LSN record that belong to each treatment phase but expected timeframe is after OEDRA/PMS1A or next phase first visit
B. There is at least one LSN eCRF."DATE OF LESION EVALUATION" for each expected timeframe according to the start/stop week and frequency from cycle 1 of each treatment phase and and with the corresponding LSN eCRF."METHOD OF ASSESSMENT" for each LSN record that belong to each treatment phase
C. Patient is not randomized
D. LSN form is empty
E. RNT.Date of Assessment is NOT NULL

SORTING

Table 1: Country, Site, Subject Number, Date of Visit, SeqTable 2: Country, Site, Subject Number, Starting Point Date, Week, Seq

2. Logic Understanding

This spec identifies, per randomized subject and per lesion, missing LSN eCRF 'DATE OF LESION EVALUATION' records under two distinct scenarios: Logic 1/Table 1 (Scenario 1) checks for a missing evaluation within a day-window around a PMS1/PMST discontinuation visit for the subject's treatment phase, while Logic 2/Table 2 (Scenario 2) checks for missing evaluations against a series of periodically-calculated 'Expected Timeframe' windows (derived from Minimal start timepoint, Start/Stop week, and Frequency) once the system date has passed each timeframe. Both logics are heavily gated by supporting-document-defined parameters (windows, method of assessment, trial segment references) and suppressed by several competing real-world events (subsequent PMS1A/OEDRA assessments, new oncologic treatment/radiation/surgery starts, next-phase overlap, RNT records, lesion merge/split reclassification) that would explain away an apparently missing evaluation. Treatment-phase scoping (via LSN Date of Initial Identification vs. phase-specific Minimal start timepoints) and numerous NA/older-vs-newer-trial toggles (PMS1A vs OEDRA, presence of merge/split item, RNT applicability) must be resolved before the core Logic 1/Logic 2 boolean conditions can be evaluated correctly.

1. Before any Logic 1/2 processing, apply the subject-level eligibility filter: the subject's data must satisfy 'Criteria Form1'.'Criteria Itemquestion1' = 'Criteria Value1' AND the same for criteria 2 and 3, all values sourced from the supporting document -- only subjects meeting all three match conditions proceed to the report logic.

2. Logic 1 (drives Table 1 / Scenario 1): First confirm there exists a 'PMS Form' (PMS1/PMST, per Note 4) record with Trial Segment Reference equal to the 'Trial Segment Reference' value from the supporting document (Logic 1.1) -- this establishes the discontinuation/reference visit exists for that treatment phase.

3. Given that PMS Form record exists, check that there is NO LSN eCRF 'DATE OF LESION EVALUATION' among the LSN records belonging to that same treatment phase that falls within a qualifying window (Logic 1.2) -- i.e., we are looking for an absence of a valid evaluation date in the window.

4. The qualifying window is defined as: LSN 'DATE OF LESION EVALUATION' >= (PMS Form DOV - Lower window of days) (Logic 1.3) AND <= (PMS Form DOV + Upper window of days) (Logic 1.4, skipped entirely if Upper window of days = NA per Note 3).

5. Additionally, any candidate LSN evaluation within that window must match on 'Method of Assessment' equal to the value defined per the corresponding 'Group for Method' in the supporting document (Logic 1.5); if Group for Method and Method of Assessment are both NA, this check is skipped entirely (Note 39).

6. Logic 1 also requires (in parentheses, i.e., an additional AND block) that there is NOT a PMS1A record with the specified 'PMS1A Trial Segment Reference' and Date of Assessment <= (PMS Form DOV - Lower window of days) (Logic 1.6), AND there is NOT an OEDRA record with Date of Assessment <= that same threshold (Logic 1.7) -- these act as suppression checks so that if a later disease assessment already occurred before the window lower bound, no missing-evaluation flag is raised. Per Note 42, if the 'Cut-off Exclude Form' parameter names PMS1A (or OEDRA), the corresponding check (1.6 or 1.7) is skipped.

7. If Logic 1.1 through 1.7 are all satisfied (PMS visit exists, no matching lesion evaluation in window, no suppressing PMS1A/OEDRA assessment), the subject/lesion/visit combination is flagged as missing and output to Table 1. *(inferred)*

8. Logic 2 (drives Table 2 / Scenario 2) begins by requiring the current System date is greater than the 'Expected Timeframe' being evaluated (Logic 2.1) -- meaning the expected timeframe window has already fully elapsed (using the maximum/end date of the range per Note 26) before it can be flagged as missing.

9. Check that there is NOT at least one LSN record in the add-entry section where the LSN 'DATE OF LESION EVALUATION' falls within the 'Expected Timeframe' for that treatment phase (Logic 2.3) AND the LSN 'METHOD OF ASSESSMENT' matches the value defined for the corresponding 'Group for Method' (Logic 2.4, using the maximum date of the range per Note 26; skipped if Group for Method/Method of Assessment = NA per Note 39) -- combined this is Logic 2.2, the absence-of-a-valid-evaluation check for this timeframe.

10. An additional lesion-merge/split condition must hold (parenthesized OR block): either the LSN 'LESION NUMBER' is not referenced in any other record's 'INDICATE IF LESION MERGED OR SPLIT' field (Logic 2.5), OR it is referenced but the Expected Timeframe is before that record's 'DATE OF LESION EVALUATION' and the merge/split indicator is NULL (Logic 2.6 + 2.7), OR the merge/split indicator is NOT NULL and the missing evaluation date is after the LSN 'DATE OF

INITIAL IDENTIFICATION' (Logic 2.8). Per Note 22, this entire merge/split sub-check is bypassed for older trials where the item doesn't exist.

11. Check that there is NOT at least one 'Oncologic Treatment' (ODBF) record where the Start Date of Regimen/Medication is not null and Expected Timeframe precedes the minimum such start date across all records (Logic 2.9, using minimum date of the range per Note 26) -- this suppresses the missing flag if a new oncologic treatment regimen started before the expected timeframe. Skipped entirely if 'Oncologic Treatment' = NA in supporting doc (Note 15).

12. Similarly check there is NOT at least one 'Oncologic Radiation' (ORAD2) record with a non-null Date where Reason is not Palliative, and Expected Timeframe precedes the minimum such ORAD2 date (Logic 2.10) -- suppresses flag if radiation treatment (non-palliative) started before the expected timeframe. Skipped if 'Oncologic Radiation' = NA (Note 16).

13. Similarly check there is NOT at least one 'Oncologic Surgery' (OSOF) record with a non-null Date of Surgery where Expected Timeframe precedes the minimum such surgery date (Logic 2.11) -- suppresses flag if surgery occurred before the expected timeframe. Skipped if 'Oncologic Surgery' = NA (Note 17).

14. Check the parenthesized OR block: there is NOT a PMS1A record (with the specified 'PMS1A Trial Segment Reference') with Date of Assessment <= Expected Timeframe (Logic 2.12), OR there is NOT an OEDRA record with Date of Assessment <= Expected Timeframe (Logic 2.13) -- using the maximum date of the timeframe range for 2.13 per Note 26, minimum for 2.12. This mirrors the Logic 1.6/1.7 suppression concept but applied against the Expected Timeframe rather than the PMS window. Skippable per Cut-off Exclude Form parameter (Note 42).

15. Check another parenthesized OR block regarding phase transition: either the subsequent treatment phase's 'Minimal start timepoint' (its DOV or Date of Assessment of SR) is NULL (Logic 2.14, meaning no next phase has started yet), OR the current 'Expected Timeframe' is still less than that subsequent phase's Minimal start timepoint (Logic 2.15) -- this prevents flagging a missing evaluation for phase X if the Expected Timeframe actually falls into or after the next phase's start (consistent with Note 27's overlap rule).

16. Finally, check there is NOT at least one RNT record with RNT.Date of Assessment not null (Logic 2.16) -- another suppression check, applicable only to newer trials since RNT is not implemented in old trials (Note 41).

17. If Logic 2.1 through 2.16 are all satisfied simultaneously (timeframe elapsed, no matching evaluation found, no merge/split disqualification, no earlier oncologic treatment/radiation/surgery start, no earlier PMS1A/OEDRA assessment, no next-phase overlap, no RNT record), that specific Expected Timeframe/lesion combination is flagged as missing and output to Table 2. *(inferred)*

18. Before applying Logic 1 or Logic 2, treatment phases must first be delineated per Note 30 using LSN 'DATE OF INITIAL IDENTIFICATION' compared against each phase's 'Minimal start timepoint' (e.g., Cycle 1, Second Course Cycle 1) combined with Tumor Identification conditions (target/non-target vs new lesion) and presence/absence of the merge/split indicator -- this scoping determines which LSN records 'belong to the corresponding treatment phase' referenced throughout Logic 1.2 and Logic 2.3/2.9-2.16. *(inferred)*

19. The 'Expected Timeframe' values used throughout Logic 2 (and referenced in Table 1 column 9/Table 2 column 10) are pre-calculated per Note 25's stepwise formula: compute Timepoint N = Minimal start timepoint + Start week + (N-1)*Frequency for each N, retain only those Timepoints between (Minimal start timepoint + Start week) and (Minimal start timepoint + Stop week), then expand each retained Timepoint by Lower/Upper window of days to form the displayed range 'From X to Y'.

20. Special handling per Note 24 overrides both Logic 1 and Logic 2 outputs: even if all other conditions are met, do not output a missing evaluation if the Expected Timeframe (or window) falls before the LSN 'DATE OF INITIAL IDENTIFICATION' for that lesion, since Expected Timeframe should always logically follow initial identification (accounts for lesions newly identified after cycle 1).

3. Spec Review

Readiness: Needs revision — The spec has strong edge-case coverage (Notes 1-42) but contains one blocking ambiguity around how the discontinuation visit is identified in Logic 1, plus several warning-level ambiguities in Logic 2's boolean grouping, undefined terms, and traceability gaps between Notes/Scenarios and explicit Logic conditions (randomization, empty form). These should be resolved before query generation to avoid guesswork on record inclusion.

BLOCKING (ambiguity): Logic 1's 'PMS Form' record with Trial Segment Reference = 'Trial Segment Reference' (Logic 1.1) is described as the discontinuation visit trigger per Scenario 1 ('expected window of days from discontinuation of that treatment phase'), but the Logic section never states that 'PMS Form' in Logic 1.1/1.3/1.4 specifically represents a discontinuation visit. There's no explicit mapping of which PMS1/PMST visit qualifies as 'discontinuation' vs any other PMS1/PMST visit with the same Trial Segment Reference. → Add an explicit statement in Logic 1 (or a Note) clarifying that 'PMS Form' in this context refers to the discontinuation/end-of-phase visit, and specify how that visit is identified (e.g., a specific visit mnemonic, form field, or flag) distinct from other PMS1/PMST visits sharing the same Trial Segment Reference.

WARNING (ambiguity): Logic 2.6/2.7/2.8 chain reads: 'LESION NUMBER is present in other ... but Expected Timeframe is before to DATE OF LESION EVALUATION AND INDICATE IF LESION MERGED OR SPLIT is NULL OR INDICATE IF LESION MERGED OR SPLIT is NOT NULL AND missing Date is after DATE OF INITIAL IDENTIFICATION'. The operator precedence between the AND/OR chain across 2.6, 2.7, 2.8 is not parenthesized, so it's unclear whether 2.7 belongs to the 2.6 clause or is an independent OR branch alongside 2.8. → Add explicit parentheses to Logic 2.6-2.8 showing intended grouping, e.g. '(2.6 AND 2.7) OR (2.6 AND 2.8)' or whatever is intended, so the boolean logic is unambiguous.

WARNING (ambiguity): Note 23 states that for Scenario 2, 'stop week' can be NA and the report will 'continue output missing lesion evaluations until PMS1A/OEDRA Date of Assessment is less than Expected Timeframe' -- but doesn't specify an upper bound on how many Timepoints/N to generate before that condition is checked, given Step B in Note 25 requires 'Stop week' to bound Timepoint N. → Clarify how Timepoint N is generated/bounded when Stop week = NA (e.g., state that system date or a maximum lookback is used to cap the number of generated timepoints before the PMS1A/OEDRA check is applied).

WARNING (ambiguity): The term 'missing Date' in Logic 2.8 ('missing Date is after the LSN eCRF.DATE OF INITIAL IDENTIFICATION') is not defined anywhere -- it's unclear whether this refers to the Expected Timeframe, a system date, or some other computed missing-evaluation date. → Define 'missing Date' explicitly in Logic 2.8, e.g. state it equals 'Expected Timeframe' or another specific computed value, consistent with usage elsewhere in Logic 2.

WARNING (consistency): Table 1 Column Heading item 6 is 'Form' and its Forms/Items mapping is 'PMS1/PMST.Form Mnemonic', while Table 2 column 7 'Form' maps to 'Lesion Evaluation Form.Form Mnemonic'. The same column name 'Form' is populated from two conceptually different source forms (visit form vs. lesion form) across the two tables without explanation. → Rename one of the columns (e.g., Table 1 'Visit Form' vs Table 2 'Lesion Form') or add a note clarifying that 'Form' refers to different source forms per table to avoid confusion during generation and review.

INFO (consistency): The spec alternates between 'Lesion Identification' and 'Tumor Identification' (Column heading 8 in Table 1: 'Lesion Identification'; Forms/Items 8: 'Lesion Evaluation Form.Lesion Identification/Tumor Identification'), and Note 30 separately references 'LSN eCRF. TUMOR IDENTIFICATION' as a distinct field used for phase determination logic. It's unclear if 'Lesion Identification' (displayed) and 'TUMOR IDENTIFICATION' (used in Note 30 conditions) are the same field or two different items. → Clarify whether 'Lesion Identification' (output column) and 'Tumor Identification' (used in Condition 1-4 of Note 30) refer to the same eCRF item, and standardize the naming across the Column Heading, Forms/Items, and Notes sections.

WARNING (completeness): Table 2 columns 14-16 (Min cut off form, Min cut off seq, Min cut off date) are populated 'For Logic 2.9~2.13' per Notes 33-35, but Logic 2.9, 2.10, 2.11 each reference three separate source forms (Oncologic Treatment, Radiation, Surgery) plus 2.12/2.13 reference PMS1A/OEDRA -- five distinct sources. The spec doesn't state how a single 'min cut off form/seq/date' triple is chosen when multiple of these logic branches (2.9-2.13) could each independently produce a qualifying minimum date. → Add explicit tie-breaking logic for Min cut off form/seq/date when multiple sources (Oncologic Treatment, Radiation, Surgery, PMS1A, OEDRA) have candidate

minimum dates -- e.g., 'use overall minimum across all applicable sources; if tie, prioritize in order Treatment > Radiation > Surgery > PMS1A > OEDRA'.

WARNING (completeness): Note 12 states 'Report will work when LSN eCRF has been initialized. If the form is empty, then, no output is produced' and this is reflected in Scenario 1 Output No (D) and Scenario 2 Output No (D) as 'LSN form is empty'. However, the Logic section (Logic 1, Logic 2) never explicitly states a check/condition for 'LSN form is empty' -- there's no corresponding Logic step for this scenario condition. → Add an explicit Logic condition (e.g., Logic 1.0 / Logic 2.0) stating 'LSN eCRF has at least one record for the subject' as a precondition, so the empty-form exclusion in Scenario Output No is traceable to Logic.

WARNING (completeness): Scenario 1 Output No (C) states 'Patient is not randomized' and Note 11 says 'Report will work for randomized patients only', but no Logic step (Logic 1.x) explicitly encodes a randomization check/date filter. → Add an explicit Logic condition referencing randomization status/date (e.g., 'Subject has a randomization date present') so the non-randomized exclusion is traceable from Logic rather than only from a Note and Scenario list.

WARNING (testability): Scenario 2 Output No (A) describes an exclusion where 'expected timeframe is after OEDRA/PMS1A or next phase first visit' -- this appears to correspond to Logic 2.12/2.13 and 2.14/2.15, but Output No (B) describes the case where an evaluation record IS found, which corresponds to Logic 2.2-2.8. Since Output No (A) and Output Yes both hinge on 'no LSN evaluation found', the distinguishing factor is buried in the combination of 2.12-2.15, making it hard to verify from Scenario text alone which specific Logic sub-conditions map to No-A vs Yes. → Add explicit references to Logic sub-numbers within each Scenario Output bullet (e.g., 'per Logic 2.12/2.13/2.14/2.15') so each scenario branch can be directly traced and tested against specific logic conditions.

INFO (naming): Forms/Items uses inconsistent capitalization/spacing for the same conceptual references, e.g. 'PMS1/PMST.Visit Mnemonic' vs 'First Visitmnemonic of Treatment Phase.Visit Mnemonic' (missing space, inconsistent casing) and 'OSOF eCRF." "DATE OF SURGERY/ DATE OF SURGERY / DATE OF SURGERY' contains a malformed/duplicated field name with stray quotes. → Clean up naming in Forms/Items and Logic 2.11 to consistently use 'Form.Item' notation (e.g., 'First Visit Mnemonic of Treatment Phase.Visit Mnemonic') and fix the malformed OSOF date field reference to a single clear item name.

INFO (completeness): Note 24 says the report should not output a missing evaluation if 'Expected Timeframe' is before 'DATE OF INITIAL IDENTIFICATION', addressing new lesions appearing after cycle 1. This is a good edge case coverage, but there's no similar explicit statement for Table 1/Logic 1 regarding new lesions appearing after the discontinuation visit window (only Logic 2/Table 2 is addressed). → Clarify whether Logic 1 (Table 1) also needs to exclude lesions whose DATE OF INITIAL IDENTIFICATION is after the discontinuation visit window, similar to Note 24 for Logic 2.

4. Generated Code

CQL — Guardrails: passed

```
SELECT
  `@HDR`.`Site`.`Country` AS `Country`,
  `@HDR`.`Site`.`Number` AS `Site`,
  `@HDR`.`Subject`.`Name` AS `Subject Number`,
  PMS1_DOV.VisitMnemonic AS `Visit`,
  DATE_FORMAT(PMS1_DOV.VisitDateRaw, '%d%b%Y') AS `Date of Visit`,
  `@Form`.`Name` AS `Form`,
  LOEL.LSN_SEQ AS `Seq`,
  LOEL.LesionId AS `Lesion Identification`,
  CONCAT(DATE_FORMAT(LOEL.ExpectedStartRaw, '%d%b%Y'), ' to ', DATE_FORMAT(LOEL.ExpectedEndRaw,
'%d%b%Y')) AS `Expected Timeframe`,
  LOEL.CriteriaValue1 AS `Criteria Value1`,
  LOEL.CriteriaValue2 AS `Criteria Value2`,
  LOEL.CriteriaValue3 AS `Criteria Value3`
FROM
  (
    SELECT
      `@HDR`.`Subject`.`ID` AS SubjectID,
      DOVFM.DOVPHASEIT AS TreatmentPhase,
      PMS1FM.PMS1DSREFIDIT AS TrialSegmentReference,
      DOVFM.DOV AS VisitDateRaw,
      DOVFM.DOVPERIODIT AS VisitMnemonic
    FROM `EDC`.`PMS1FM` PMS1FM
    INNER JOIN `EDC`.`DOVFM` DOVFM
      ON `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DOVFM.DOVSERRORVIT = 'N' OR DOVFM.DOVSERRORVIT IS NULL)
  ) PMS1_DOV
  INNER JOIN
  (
    SELECT
      `@HDR`.`Subject`.`ID` AS SubjectID,
      LOELFM.LOELSPIDIT AS LSN_SEQ,
      LOELFM.LOELTUALSNNUM1TB AS LesionId,
      RAWDATE(LOELFM.LOELLSDTCLEVIT) AS LesionEvalDateRaw,
      LOELFM.LOELLSMETHODIT AS MethodOfAssessment,
      RAWDATE(LOELFM.LOELTUDTCIT) AS InitialIdentificationDateRaw,
      SUBDATE(RAWDATE(LOELFM.LOELTUDTCIT), INTERVAL -0 DAY) AS ExpectedStartRaw,
      RAWDATE(LOELFM.LOELTUDTCIT) AS ExpectedEndRaw,
      '<UNKNOWN>' AS CriteriaValue1,
      '<UNKNOWN>' AS CriteriaValue2,
      '<UNKNOWN>' AS CriteriaValue3
    FROM `EDC`.`LOELFM` LOELFM
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (LOELFM.DELETEDFM = 'N' OR LOELFM.DELETEDFM IS NULL)
  ) LOEL
  ON PMS1_DOV.SubjectID = LOEL.SubjectID
WHERE
  `@HDR`.`Subject`.`Name` LIKE '%--%'
  AND LOEL.LesionEvalDateRaw IS NULL
  AND NOT EXISTS (
    SELECT 1 FROM `EDC`.`PMS1FM` P1A
    WHERE `@HDR`.`Subject`.`ID` = PMS1_DOV.SubjectID
      AND `@Form`.`Status` = 'submitted__v'
      AND RAWDATE(P1A.PMS1DSDTCIT) <= SUBDATE(PMS1_DOV.VisitDateRaw, INTERVAL 7 DAY)
  )
  AND NOT EXISTS (
    SELECT 1 FROM `EDC`.`OEDRAFM` OED
    WHERE `@HDR`.`Subject`.`ID` = PMS1_DOV.SubjectID
      AND `@Form`.`Status` = 'submitted__v'
      AND RAWDATE(OED.OEDRADSDTCIT) <= SUBDATE(PMS1_DOV.VisitDateRaw, INTERVAL 7 DAY)
  )
ORDER BY
  `Country`, `Site`, `Subject Number`, `Date of Visit`, `Seq`
```

This is an extremely complex, supporting-document-driven spec (external CSV with per-trial Criteria Forms/Items, window days, phase definitions, method-of-assessment mappings, etc.) that cannot be fully parameterized in static

CQL since the actual Criteria Form/Item names, window days, frequencies, and phase definitions are only known at runtime from an uploaded supporting document -- I've built Table 1's core Logic 1 shape (PMS1/DOV visit join to LOEL lesion records missing DATE OF LESION EVALUATION within the window, excluding PMS1A/OEDRA assessments before the window per Logic 1.6/1.7) using LOELFM as the Lesion Evaluation Form (LOELLSDTCLEVIT = date of lesion evaluation, LOELTUDTCIT = date of initial identification, LOELTUALSNNUM1TB = lesion identification, LOELLSMETHODIT = method of assessment, LOELSPIDIT = seq) and PMS1FM/DOVFM for the PMS visit and DOV date, filtering deleted (DELETEDFM) and erroneous visits (DOVSERRORVIT) per Notes 12/13, and randomized-only via the confirmed Subject.Name LIKE '%--%' pattern. However, Criteria Value1/2/3 columns, the exact window-day/frequency-driven Expected Timeframe calculation (Note 25), Table 2's Expected-Timeframe-based Logic 2 (with Oncologic Treatment/Radiation/Surgery/RNT exclusions, min-cutoff form/seq/date, and subsequent-phase overlap logic), and the Cut-off Exclude Form parameter cannot be hardcoded since they depend entirely on the runtime supporting-document CSV values (Trial Segment Reference, Lower/Upper window, Method of Assessment, Start/Stop week, Frequency) which have no fixed Form/Item mapping -- these are flagged as major unresolved assumptions requiring the actual supporting document content and DESCRIBE FORM verification of LOELFM/PMS1FM/DOVFM/OEDRAFM/PMS1FM item names before this can run correctly; Table 2 was not built out due to this same limitation, and the report should also implement the explicit "Supporting Document Not Present" and zero-result messaging per Notes 28/zero-result convention.